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Breast pump accessory system

Abstract

A pumping top may include a left panel and a right panel, each of the left panel and the right panel having a bottom edge and being configured to cover a side of a wearer's torso and cover and support a breast of the wearer. Each of the left panel and the right panel may include (i) an expandable bladder configured to contact and partially surround a corresponding breast of the wearer, (ii) a supply line coupled at one end to the bladder and at an opposite end to a connector disposed proximate the bottom edge, and (iii) a return line coupled at one end to the bladder and at an opposite end to the connector. The connector may be configured to couple to a circulating pump that is separate and distinct from the pumping top, which circulates warmed liquid through the pumping top.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation-in-part of U.S. patent application Ser. No. 18/217,985, titled “Breast Pump Accessory System,” filed on Jul. 3, 2023, which claims the benefit of U.S. Provisional Application Ser. No. 63/358,069, titled “Breast Pump Accessory System and Methods of Use,” filed on Jul. 1, 2022. This application further claims the benefit of U.S. Provisional Application Ser. No. 63/554,396, titled “Breast Pump

Accessory System,” filed on Feb. 16, 2024. (2) This application incorporates the entire contents of the foregoing applications herein by reference.

TECHNICAL FIELD

(1) Various implementations relate generally to accessory systems for use with breast pumps.

BACKGROUND

(2) Traditional electric breast pumps that only use suction power to remove milk from a breast may only remove milk closest to the nipple of a breast—which may lead to incomplete extraction and low milk yield, sometimes despite long, uncomfortable pumping sessions. In contrast to such electric breast pumps, a nursing infant uses a combination of suction, warmth (e.g., from its mouth and hands), and movement (e.g., from its jaw and hands against the breast) to efficiently remove milk from the breast.

SUMMARY

(3) A pumping top may include a left panel and a right panel, each of the left panel and the right panel having a bottom edge and being configured to cover a side of the wearer's torso and cover and support a breast of a wearer. The pumping top may further include a zipper configured to removably couple the left front panel and the right front panel. Each of the left panel and the right panel may include (i) an expandable bladder configured to contact and partially surround a corresponding breast of the wearer, (ii) a supply line coupled at one end to the bladder and at an opposite end to a connector disposed proximate the bottom edge, and (iii) a return line coupled at one end to the bladder and at an opposite end to the connector. The supply line and the return line may be disposed in the corresponding left or right panel. Each of the left panel and the right panel may have an aperture configured to receive an external breast pump that is separate and distinct from the pumping top. The connector may be configured to couple to a circulating pump that is separate and distinct from the pumping top, which circulating pump circulates warmed liquid to each supply line and receives return liquid from each return line.

(4) In some implementations, each of the right and left panels comprise inner and outer layers, and the supply and return lines are disposed therebetween. Each connector may further include a valve to seal off the supply line and the return line when they are not coupled to the circulating pump. The cross-sectional shapes of the inlet line and the outlet line may be flat, with a width dimension being greater than a height dimension. The inlet and outlet lines may be configured to be disposed along the side of the wearer.

(5) In some implementations, the expandable bladder of each side comprises a first expandable bladder and a second expandable bladder. The first expandable bladder may be coupled to the supply line, the second expandable bladder may be coupled to the return line, and the first expandable bladder may be coupled to the second expandable bladder with a tube. The first expandable bladder and the second expandable bladder may be expandable, and the tube may be non-expandable. The tube may have a reduced diameter relative to the first expandable bladder and the second expandable bladder, such that fluid flow from the first expandable bladder to the second expandable bladder is slowed to cause the first expandable bladder to fill prior to the filling of the second expandable bladder.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 illustrates a breast pump accessory system, according to some implementations.

(2) FIG. 2 illustrates a front view of a brassiere of the breast pump accessory system, according to some implementations.

(3) FIG. 3 illustrates the brassiere of the breast pump accessory system, including outer and inner expandable elements, according to some implementations.

- (4) FIG. 4 illustrates a front view of the brassiere, including an inlet line and an outlet line, according to some implementations.
- (5) FIGS. 5A and 5B illustrate interior views of the brassiere, according to some implementations.
- (6) FIG. 6 illustrates an interior view of the brassiere, with another exemplary flow path.
- (7) FIGS. 7A, 7B, and 7C illustrate an expandable element in varying stages of inflation, according to some implementations.
- (8) FIG. 8 illustrates a flowchart demonstrating the use of the breast pump accessory system, according to some implementations.
- (9) FIGS. 9A and 9B illustrate an exemplary pumping top that is configured to be worn by a nursing user to facilitate expression of breast milk.
- (10) FIGS. 9C and 9D illustrate exemplary implementations that provide a warming function independently of a massage function.
- (11) FIGS. 9E and 9F illustrate details of exemplary mating connectors to couple the pumping top to a circulating pump.
- (12) FIG. 9G is a cross-sectional view showing exemplary disposition of supply and return lines.
- (13) FIGS. 10A-10F depict operation of exemplary expandable bladders.
- (14) FIGS. 11A-11F depict operation of an exemplary system having first and second bladders with a flow restrictor therebetween.
- (15) FIGS. 12A-12F depict operation of another exemplary system having first and second bladders with a flow restrictor therebetween and a check valve.
- (16) FIGS. 13A-13F depict operation of another exemplary system having first and second bladders with a flow restrictor therebetween and a check valve, wherein a negative pressure is applied to deflate the bladders.
- (17) FIGS. 14A-14F depict operation of another exemplary system having first and second bladders with a flow restrictor therebetween and two check valves.
- (18) FIGS. 15A-15F depict operation of another exemplary system having first and second bladders with a flow restrictor therebetween and two check valves in different positions.
- (19) FIGS. 16A-16C illustrate one implementation in which bladders are prevented from overinflating.
- (20) FIGS. 17A-17C illustrate another implementation in which bladders are prevented from overinflating.
- (21) FIG. 18A illustrates an exemplary stress-strain curve that depicts the staged inflation of first and second bladders at different pressures.
- (22) FIG. 18B illustrates an exemplary time-strain curve that depicts the staged inflation of first and second bladders at different times.

DETAILED DESCRIPTION

- (23) FIG. 1 illustrates a breast pump accessory system **100**. In some implementations, as shown in FIG. 1, the system **100** includes a fluid reservoir **102** configured to store fluid **104**, a heating element **106** coupled to the fluid reservoir **102**, and configured to heat the fluid **104**, an inlet line **108**, and an outlet line **110**. A pump **116** is configured to transfer fluid **104** to a brassiere **112** via an inlet line **108** and receive fluid **104** back from the brassiere **112** via an outlet line **110**.
- (24) In some implementations, the brassiere **112** comprises a plurality of expandable elements **114** that are in fluid communication with the inlet line **108** and the outlet line **110**. And the fluid **104** circulated by the pump **116** may be used to inflate and deflate the plurality of expandable elements **114** to provide positive pressure, or massage, to each breast of a user of the system **100**. Further, warming the fluid **104** via the heating element **106** can provide heat therapy to each breast of the user. In some implementations, the combination of massage and heat may help facilitate breast milk expression, in combination with the use of an external breast pump (not shown).
- (25) In some implementations, the pump **116** comprises a flow control mechanism configured to regulate the amount of fluid **104** allowed to flow through the inlet line **108**. The pump **116** may

comprise a flow rate control mechanism configured to regulate the speed at which the fluid **104** is flowed through the inlet line **108**, expandable elements **114** of the brassiere **112** and back through the outlet line **110** (collectively, the circulation system **115**). Adjustments to the volume or speed of the fluid **104** flow through the circulation system **115** may implement variations in massaging force (i.e., positive pressure) and speed effectuated by inflation of the plurality of expandable elements **114**.

(26) FIG. 2 illustrates an exemplary front view of the brassiere **112**. Similar to traditional “pumping bras,” the brassiere **112** may include a plurality of holes **206**, wherein each hole is configured to receive a breast pump flange—also commonly referred to as a nipple flange, nipple shield, or breast shield—to facilitate the use of a traditional breast pump. In some implementations (not shown), the system **100** includes a plurality of silicone nipple flanges, or breast shields, configured to be received by the plurality of holes **206** and couplable to an external breast pump (not shown). In such implementations, use of a soft and flexible material, such as silicone, rather than the traditional hard plastic, may enable the flanges/shields to sustain contact with the user's breasts during a pumping session involving suction and massage. For example, a flexible flange/shield may better move with the user's breast as the massaging pressure is applied than a rigid plastic flange/shield.

(27) As shown, the brassiere **112** includes two straps **200** and a band **202**, all of which may be a “wide fit” for added comfort for the user (e.g., as opposed to narrow straps and/or a narrow band, which may “dig in” to the user's shoulders and/or ribcage and may cause discomfort). As shown in FIG. 2, the brassiere **112** may also include a front closure via a zipper **204**, which may improve the ease of use of the brassiere **112**.

(28) FIG. 3 illustrates additional detail of exemplary expandable elements **114**. As shown, the brassiere **112** includes a first outer expandable element **300a** and a first inner expandable element **302a** on one side, and a second outer expandable element **300b** and a second inner expandable element **302b** on the other side. Each of the first outer expandable element **300a**, the second outer expandable element **300b**, the first inner expandable element **302a**, and the second inner expandable element **302b** may define a C-shape **304**, as illustrated. In some implementations, the C-shape **304** is sized and configured such that when the user is wearing the brassiere **112**, the plurality of expandable elements **114** is configured to at least partially surround an outer portion of each breast of the user. In other implementations, the expandable elements **114** may be configured in another manner (e.g., in a horseshoe shape or inverted horseshoe shape).

(29) In some implementations, the sizes of the expandable elements **114** may be configured to correspond to the size of the brassiere **112**. For example, a smaller brassiere **112** may include smaller expandable elements **114** than those expandable elements **114** in a larger brassiere **112**.

(30) In some implementations, the expandable elements **114** may be differently sized relative to each other. For example, as shown in FIG. 3, the first and second outer expandable elements **300a**, **300b** are shown as larger than the first and second inner expandable elements **302a**, **302b**. In some implementations, this is a function of the their position relative to a user's breasts—the outer expandable elements **300a**, **300b** may be configured to contact the base of a user's breasts, whereas the inner expandable elements **302a**, **302b** may be configured to contact a smaller, more distal portion of the user's breasts; in some implementations, different sizes of the expandable elements **114** relative to each other can facilitate specific massage patterns (e.g., massage from the base of a user's breasts toward the nipples, to facilitate milk expression).

(31) Different numbers of expandable elements **114** may be included in the brassiere **112**. For example, two expandable elements on each side, **300a** and **302a**, and **300b** and **302b**, are shown in FIG. 3; whereas three expandable elements are depicted on each side in FIG. 1. In general, the plurality of expandable elements **114** may include one, two, three, four, or more than four expandable elements on each side of the brassiere **112**.

(32) FIG. 4 provides another front view of the brassiere **112**, showing an exemplary termination for

the inlet line **108** and outlet line **110**. As described above, fluid **104** may be circulated into inlet line **108**, through the brassiere **112**, and out of the outlet line **110**. As shown in FIG. 4, inlet line **108** is shown as terminating on one side of the zipper **204**, while the outlet line **110** is shown as terminating on the other side of the zipper **204**. In other implementations, the inlet line **108** and the outlet line **110** may be located adjacent to one another and may be on the same side of the zipper **204**. In other implementations, the inlet line **108** and outlet line **110** may be terminated in another location on the brassiere **112**. In still other implementations, dedicated inlet and outlet lines may be provided to each side of the brassiere **112**.

(33) FIG. 5A illustrates an example flow path of the fluid **104** through the inlet line **108**, each expandable element **114**, and out the outlet line **110**. The flow path shown in FIG. 5 is intended as a non-limiting example of how fluid **104** may flow through the brassiere **112**. Other different flow paths are possible to achieve the desired outcome of applying massage and/or heat to a user's breast to improve the expression of breast milk.

(34) As illustrated by the dashed arrows in the implementation shown in FIG. 5A, the fluid **104** enters the brassiere **112** via the inlet line **108**. In some implementations, at least one of the inlet line **108** and the outlet line **110** are located between fabric layers of the brassiere **112**. In some implementations, at least one of the inlet line **108** and the outlet line **110** may be located along an interior portion of the brassiere **112**. In some implementations, at least one of the inlet line **108** and the outlet line **110** may be located along an exterior portion of the brassiere **112**.

(35) In some implementations, as shown, the inlet line **108** is configured to split into two lines, to transfer fluid **104** into the first outer expandable element **300a** and the second outer expandable element **300b**. After flowing up through the first and second outer expandable elements **300a**, **300b**, the fluid **104** may be configured to flow down through the first and second inner expandable elements **302a**, **302b** into separate segments of the outlet line **110**. As illustrated in FIG. 5A, the segments of the outlet line **110** may converge into a single outlet line **110** to direct fluid **104** out of the brassiere **112**. In some implementations, rather than a single line that splits off, the system **100** may include more than one inlet line **108** and/or more than one outlet line **110**. For example, each of the first and second outer expandable elements **300a**, **300b** may be coupled to an independent inlet line **108**. Similarly, each of the first and second inner expandable elements **302a**, **302b** may be coupled to an independent outlet line **110**. In such implementations, the brassiere **112** may be coupled to four separate lines for transferring the fluid **104** between the fluid reservoir **102** and the brassiere **112**.

(36) FIG. 5B provides additional detail for exemplary expandable elements **114**, according to one implementation. In implementations in which the inlet line **108** splits into separate segments (e.g., as shown in FIG. 5A), one segment may couple to a bottom portion **600a** of the first outer expandable element **300a** and one may couple to a bottom portion **600b** of the second outer expandable element **300b**. Similarly, the outlet line **110** may comprise separate segments, one of which may couple to a bottom portion **602a** of the first inner expandable element **302a** and one of which may couple to a bottom portion **602b** of the second inner expandable element **302b**.

(37) In some implementations, when fluid **104** moves up the first outer expandable element **300a** from the bottom portion **600a** to the top portion **606a**, it reaches a first tube **604a**, as shown in FIG. 5B. The first tube **604a** may comprise a non-expandable material and be coupled to the top portion **606a** of the first outer expandable element **300a** and the top portion **608a** of the first inner expandable element **302a**, such that the fluid **104** transfer from first outer expandable element **300a** to the first inner expandable element **302a**, via the first tube **604**. In some implementations, in addition to the first tube **604a** being non-expandable (such that flow of fluid **104** through the first tube **604a** does not cause any expansion, and thus does not effectuate pressure on the user at its location), the first tube **604a** also has a reduced diameter relative to the expandable elements **300a** and **302a**. This reduced diameter can slow the transfer of the fluid **104** from the first outer expandable element **300a** to the first inner expandable element **302a**, to, in some implementations,

facilitate an outer-to-inner massage force on a wearer's breast.

(38) In some implementations, the pump **116** (see FIG. **1**) may be configured to cause fluid **104** to flow through the system **100** substantially continuously while the system **100** is in use; in other implementations, the pump **116** may be configured to cause fluid **104** to intermittently flow through the system **100** (e.g., to implement a rhythmic massage function). A control system for the pump **116** may facilitate user adjustment of the frequency of intermittent flow, as well temperature, volume and/or flow of the fluid **104** (e.g., to vary the massage force and other parameters).

(39) One flow path has been described, but other flow paths are possible by varying the orientation and shape of the outer expandable elements **300a** and **300b**, inner expandable elements **302a** and **302b**, first and second tubes **604a** and **604b**, and orientation of the inlet line(s) **108** and outlet line(s) **110**. Thus, fluid **104** may flow down through the first and second outer expandable elements **300a**, **300b**, and up through the first and second inner expandable elements **302a**, **302b**; or it may flow in another manner.

(40) In some implementations, the flow path of the fluid **104**, as illustrated in FIGS. **5A** and **5B** through the C-shape **304** of the plurality of expandable elements **114**, is configured to provide positive pressure to mimic massage patterns recommended by lactation consultants (e.g., a C-shape motion in an outward-in pattern, from the base of each breast toward the tip, or nipple). The filling of the first and second outer expandable elements **300a**, **300b**, followed by the filling of the first and second inner expandable elements **302a**, **302b** may be configured to provide massage of each breast in an outward-in pattern. Further, the plurality of expandable elements **114** may be configured to fill and empty in a sequential pattern. For example, the first and second outer expandable elements **300a**, **300b** may be configured to fill, then empty as the first and second inner expandable elements **302a**, **302b** fill with the fluid **104**.

(41) Several flow patterns may be possible with the system **100**. In some implementations, the first and second outer expandable elements **300a**, **300b** are configured to fill substantially simultaneously. Similarly, the first and second inner expandable elements **302a**, **302b** may be configured to fill substantially simultaneously. In some implementations, the first outer expandable element **300a** is configured to at least partially fill before the second outer expandable element **300b** (or vice-versa), and the first inner expandable element **302a** is configured to at least partially fill before the second inner expandable element **302b** (or vice-versa). In some implementations, the system **100** is configured such that the fluid **104** is configured to flow into only one side of the brassiere **112** at a time. In such implementations, a user may be able to nurse an infant from one breast while simultaneously pumping from the other breast. In other implementations (e.g., such as the one illustrated in FIG. **6**—viewed from the back/inside), inlet/outlet ports **109a** and **109b** may be provided for each side of a pumping top.

(42) Referring now to FIGS. **7A**, **7B**, and **7C**, an expandable element **700** is shown in varying states of inflation or expansion. As shown, the expandable element **700** can depict any of the expandable elements **114**, including the first outer expandable element **300a**, the first inner expandable element **302a**, the second outer expandable element **300b**, and the second inner expandable element **302b**. In some implementations, the system **100** is configured such that a user (e.g., through a flow control mechanism) may adjust the amount of fluid **104** flowing through the plurality of expandable elements **114**, thereby adjusting the volume and/or pressure (and thus massage intensity) applied to each breast. FIG. **7A** illustrates an expandable element **700** in a deflated state **702**, while FIG. **7C** illustrates the expandable element **700** in a fully inflated state **706**. FIG. **7B** shows the expandable element **700** in a partially inflated state **704** and is intended to represent any level of “partial inflation” possible.

(43) The system **100** may also facilitate user control of temperature. In some implementations, the heating element **106** includes an adjustment mechanism configured to enable users to adjust the temperature of the fluid **104** to suit their preference and/or needs. Users may find that different temperatures are desirable for different goals (e.g., relaxation, pain relief, or letdown promotion).

In some implementations, the heating element **106** can be entirely turned off such that the fluid **104** is not heated before/after flowing through the system **100**. This feature may be desirable, for example, when a user uses the system **100** on a hot day or inside a hot building. In some implementations, a cooling element is also provided to facilitate therapeutically cooling temperatures (e.g., to reduce swelling or pain). In other implementations, no cooling element may be provided, but the system **100** may be configured to allow a user to add ice or other cooling agents to circulating fluid. The temperature range of the fluid **104** may be configurable at least between 50 and 120 degrees Fahrenheit, but in some implementations, a greater range may be possible (e.g., below 50 degrees, to close to 32 degrees; or above 120 degrees, to 130 degrees or greater (e.g., to facilitate initial heating of the bladders and lines in the system, at which point, temperature may be reduced)).

(44) A system **100** that enables adjustment of both the massage intensity and the temperature can facilitate a highly customizable pumping session. For example, a user can vary the massage intensity and temperature (and, separately, the suction force of an external breast pump) to find her optimal combination. Combining massage, heat, or both with suction may also reduce the need to “max out” the suction force in an effort to maximize breast milk expression, resulting in a more comfortable pumping experience for the user. Further, adding massage and/or heat to the suction force provided by a traditional breast pump may increase the speed of breast milk expression, resulting in shorter pumping sessions for the user.

(45) FIG. **8** illustrates an exemplary method for using the system **100**. In some implementations, the method starts with a user filling (**800**) the fluid reservoir (e.g., reservoir **102**, shown in FIG. **1**). The fluid reservoir **102** may be sized and configured for portability (e.g., about the size of a tea kettle or smaller, in some implementations). The method includes turning on (**802**) the heating element **106**. As previously discussed, the heating element **106** may include a temperature adjustment mechanism allowing the user to set (**804**) a desired temperature. If a user does not desire to heat the fluid **104**, steps **802** and **804** may be skipped.

(46) The method includes the user putting on (**806**) the brassiere **112**. In some implementations, the brassiere **112** is configured for all-day wear; thus, the user may already have put on (**806**) the brassiere **112** prior to execution of the other steps.

(47) The user may connect (**808**) the inlet line **108** and the outlet line **110** from the fluid reservoir **102** to the brassiere **112**, and turn on (**810**) the pump **116** to prepare to initiate the flow of the fluid **104** through the inlet line **108**. The pump **116** may include a flow control mechanism and/or a flow rate control mechanism, allowing the user to set (**812**) a desired pressure setting. In some implementations, the flow rate may include a range of at least 0 to 7 L/min. After all settings have been configured, the user may initiate flow by actuating (**814**) a control (e.g., a “start” control). The above-described method is merely exemplary. In other implementations, steps may be omitted or modified; other steps may be included.

(48) The system **100** may include a control panel with various user controls including, but not limited to, a power button, a start/stop button, a time button, a pressure control button, a temperature control button, and any number of other user controls not specifically stated in this disclosure. The control panel may be fixedly coupled to an element of the system **100**, for example, the circulating pump **934**. In some implementations, the control panel is a remote device, physically disconnected from other portions of the system. For example, the system **100** may be configured to be controlled by a remote application on a remote computing device, such as a smartphone, tablet, or laptop. In some implementations, the system **100** is battery-powered (e.g., powered with direct current (DC)); in other implementations, the system **100** may run on alternating current (AC) power and require a connection to a power outlet for operation.

(49) In some implementations, the expandable elements **114** are fixedly coupled to the brassiere **112** and are sufficiently durable to withstand laundering (e.g., hand or machine washing). Similarly, at least a portion of the inlet line **108** and the outlet line **110** (e.g., the portions illustrated in FIG. **5**)

may be fixedly coupled to the brassiere **112** and may also be sufficiently durable to withstand laundering. In some implementations, at least one of the plurality of expandable elements **114**, the inlet line **108**, and the outlet line **110** are removably coupled to the brassiere **112** and are configured to be removed prior to laundering the brassiere **112**.

(50) In some implementations, the plurality of expandable elements **114** may comprise individually expandable “pockets” at least partially separated by non-compliant portions. In such implementations, the fluid **104** may take a zig-zag pattern through each expandable element rather than the up and down pattern illustrated in FIGS. **5** and **6**. Further, rather than fluid **104**, the plurality of expandable elements **114** may be configured to fill with air. Any number of methods for providing pneumatic compression, positive pressure, or other massage elements may be employed by the system **100** to facilitate breast milk expression.

(51) In some implementations, rather than pressure, the “massage” element of the system **100** may include vibration. For example, the brassiere **112** may be detachably or fixedly coupled to at least one vibrating element configured to provide sufficient movement to facilitate breast milk expression. Rather than heated fluid, the system **100** may include resistive circuits to apply heat to each breast of the user. For example, wires may be detachably or fixedly coupled to the brassiere **112** and configured to heat up in order to warm the brassiere **112**. Other methods of applying heat may be suitable for use with the system **100**.

(52) FIGS. **9A** and **9B** illustrate another implementation of an exemplary pumping top **901** that is configured to be worn by a nursing user to facilitate expression of breast milk (e.g., in the context of pumping). FIG. **9A** illustrates an exterior front of the pumping top **901**; and FIG. **9B** illustrates an interior front of the pumping top **901**. As shown, the pumping top **901** includes a right panel **904a** and a left panel **904b** (right and left from the perspective of a nursing user **919** wearing the pumping top **901**).

(53) As shown, a zipper **907** removably couples the right panel **904a** and left panel **904b**. Each of the right panel **904a** and left panel **904b** has a bottom edge, **910a** and **910b**, respectively; and in some implementations, the panels **904a** and **904b** are configured to extend some distance below the breasts **916a** and **916b** of the wearer **919** to form a tank-style or longline bra (e.g., for improved comfort). In such implementations, the pumping top **901** covers a side and front of the torso **922** of the wearer **919** and at least partially covers and supports each breast **916a** or **916b** of the wearer **919**.

(54) As shown, each of the right panel **904a** and left panel **904b** includes an aperture **913a** or **913b** that is configured to receive a breast flange associated with an external breast pump (not shown). In some implementations, each aperture **913a** or **913b** may comprise a circular opening that is configured to be aligned close to the corresponding nipple of the user **919**, and a slot that extends away from, or on either side of, the circular opening to enable a breast flange to be manipulated through the pumping top **901**.

(55) As shown in FIG. **9B**, each of the left panel **904b** and right panel **904a** includes an expandable bladder (as shown, an outer expandable bladder and inner expandable bladder are shown on each side, including right outer expandable bladder **925a** and right inner expandable bladder **928a**). In some implementations, as shown, the expandable bladders **925a** and **928a** are configured to contact (either directly or through a material layer of the pumping top **901**) a corresponding breast of the wearer **919**. A supply line **931a** supplies the expandable bladder **925a** with fluid that is circulated by a circulating pump **934**, and a return line **934a** routes fluid back to the circulating pump **934**. (As used herein, a “circulating pump” may refer to both a pump that drives fluid through a loop having a dedicated supply line **931a** and return line **934a**, or to a pump that drives fluid in and out of a blind channel (e.g., as described below with reference to FIGS. **11A-13F**); and “circulating” may refer to flow of fluid in either a loop or into and out of a channel.) In some implementations, as shown, a similar supply line, return line and one or more expandable bladders are provided with the other panel.

(56) FIG. 9C illustrates a variation of the implementation shown in FIG. 9B—specifically, an implementation in which dedicated warming loops **927a** and **927b** are provided. The warming loops may be non-expandable sections (that is, sections that do not expand and contract like the bladder sections) that may include segments **926a** and **926b** that couple the outer bladder (e.g., outer bladder **925a**) to the inner bladder (e.g., inner bladder **928a**). Such warming loops **927a** and **927b** may provide warming of a user's breasts, separate from the massage function provided by the bladders (e.g., bladders **925a** and **928a**). In some implementations, “warming loops” may provide cooling as well—that is, as used herein, “warming” may refer to a temperature regulation function that either warms or cools.

(57) FIG. 9D illustrates another implementation in which a warming loop **929** is provided that is independent of the bladders **925a** and **928a**. In such an implementation, a warming function may be provided whether or not a massage function is also provided. As shown, one connector **943a** may be provided for the bladders and massage function; another connector **944a** may be provided for the warming function; and each function (e.g., massage and warming) may be separately controllable by the circulating pump **934**.

(58) In either implementation shown in FIGS. 9C and 9D, the warming function may be more expansive than may otherwise be possible when warming and massage functions are combined. That is, warming may be provided over a greater portion of the user's breasts (e.g., over substantially 360 degrees, at least adjacent the user's nipples, and especially around the breasts in areas other than those contacted by the bladders). Moreover, the warming function may be substantially continuously provided, regardless of the frequency of any massage function or of the point in a massage cycle the bladders may be at. By disposing the warming path **929** opposite the bladders, some implementations may advantageously warm a large area between the breasts. In addition, implementations that employ non-expandable channels may occupy less space than the bladders, facilitating more surface contact for warming than the bladders.

(59) While FIGS. 9C and 9D illustrate implementations in which warming loops **927a** or **929** comprise a “channel,” or tubular construction, “patch” designs are also possible. For example, some implementations may include a thin, contoured reservoir (not shown) that can be filled with fluid and employed as a heating patch. Such a patch may be configured to provide access to the nipple via an aperture. That is, the reservoir may be segmented to enable a breast pump flange (not shown) to be disposed through such an aperture and be placed in contact with a user's nipple during a pumping operation.

(60) In some implementations, a connector **940a** may be provided at an end of the supply line **931a** and return line **931b** and may be configured to couple with a mating connector **943a** that is coupled to supply and return lines of the circulating pump **934**. In such implementations, the circulating pump **934** may be easily disconnected from the pumping top **901** and reconnected when desired. In some implementations, the connectors (e.g., connector **940a**) may be disposed proximate a bottom edge (e.g., coinciding with or close to a natural boundary for other clothing that may be worn over the pumping top **901**, such that the connector **943** may be both easily accessed and easily hidden when not in use).

(61) FIGS. 9E and 9F illustrate additional detail of exemplary mating connectors **940a** and **943a**. In FIG. 9E, the mating connectors **940a** and **943a** are shown disconnected. As shown, each connector **940a** and **943a** includes supply and return lines. One or both of the connectors (e.g., connector **940a**) may include a valve **946** (e.g., multiple flaps of material that are configured to normally abut each other to provide a seal, but that can be opened or spread to permit coupling with and sealing against another member—as depicted in FIG. 9F, in which the connectors **940a** and **943a** are shown coupled). With such a valve **946**, any liquid in the supply line **931a** or return line **937a** may be retained therein when the connectors **940a** and **943a** are decoupled, thereby preventing or minimizing leakage of any warming/inflating fluid onto the wearer or the wearer's clothing. A connecting mechanism, such as a clip **949** may be included on the connectors **940a** to **943a** to

enable them to be releasably but securely coupled.

(62) In some implementations, as shown in FIGS. **9B** and **9G**, the supply and return lines (e.g., supply line **931a** and return line **937a**) may be disposed on a side of the pumping top **901**, such that they run between the torso **922** and arm **923** of the wearer **919**. Such implementations may facilitate concealing of the lines **931a** and **937a** under other clothing of the wearer **919**. Moreover, as shown, the lines **931a** and **937a** may have a flat (e.g., elongated, with a width dimension being greater than a height dimensioned) or contoured cross-section (e.g., curved, to follow a curved surface of the side of a wearer **919**), to minimize discomfort associated with the lines **931a** and **937a** otherwise digging into the skin of the wearer **919**. To further enhance comfort of the wearer, the lines **931a** and **937a** may be disposed between two layers of material of the pumping top **901** (e.g., between an inner layer **952** and an outer layer **955**).

(63) As shown, the lines **931a** and **937a** are separate lines; but in some implementations, they may be concentrically disposed, one inside of the other. Various shapes and configurations of the lines **931a** and **937a** are possible.

(64) FIGS. **10A-10F** depict operation of exemplary expandable bladders **925a** and **928a**. FIGS. **10A-10F** are cross-sectional top views, showing material **958** of the pumping top **901** disposed against a breast **916a** of the wearer. Expandable bladder **925a** is shown in contact with a base portion of the breast **916a**, and expandable bladder **928a** is shown disposed against a more distal portion of the breast **916a** (closer to nipple **959**). Initially, both bladders **925a** and **928a** are empty (FIG. **10A**). As the circulating pump **934** is actuated, fluid fills the expandable bladder **925a**, causing it to expand and impinge on the breast **916a** (FIGS. **10B-10C**); fluid then travels into the expandable bladder **928a**, such that both bladders **925a** and **928a** contain fluid (FIGS. **10C-10D**); next, the circulating pump **934** begins evacuating the bladders **925a** and **928a**, such that bladder **925a** empties, followed by bladder **928a** (FIGS. **10E-10F**). The circulating pump **934** may cause this cycle to be repeated, thereby effectuating a massaging motion from base of the breast **916a** towards the nipple **959**. In some implementations, particularly when the fluid is warmed by the circulating pump **934**, milk expression may be enhanced.

(65) In some implementations, different portions of the expandable bladders **925a** and **928a** may comprise different materials or have different material properties. For example, portions of bladders **925a** and **928a** that contact the material **958** of the pumping top **901** may be flexible but may not stretch, whereas portions of the bladders **925a** and **928a** that are in contact with the breast **916a** may be flexible-allowing most of the distension of the bladders to be in the direction of the breast **916a**, thereby focusing the massaging force there.

(66) In some implementations, distension of the expandable bladders may be limited by the material of the bladders. For example, the bladders **925a** and **928a** may be configured to only inflate so far (e.g., with fibers, bands or other means to limit their distension), for safety reasons (e.g., to avoid too much pressure on the breast **916a**) or to maintain the integrity of the pumping top **901** (e.g., to prevent bursting of the expandable bladders **925a** and **928a**).

(67) In some implementations, the expandable bladders **925a** and **928a** and the supply and return lines **931a** and **937a** may comprise a resilient material that is configured to collapse when not subjected to pressurized fluid. Such implementations may cause fluid to be squeezed out of the expandable bladders **925a** and **928a** and the supply and return lines **931a** and **937a** when the circulating pump **934** is not actively forcing fluid therethrough-which may have the benefit of increasing comfort of the wearer when the circulating pump **934** is not in use and of minimizing any leakage when the circulating pump **934** is disconnected.

(68) FIGS. **11A-11F** depict operation of exemplary first bladder **1151** and second bladder **1152** that are arranged in a “blind channel” manner. That is, the first bladder **1151** and second bladder **1152** are arranged in series but are supplied by a single inlet/outlet **1154** at one end, through which fluid both enters and exits. As shown, a flow restrictor **1163** couples the first bladder **1151** to the second bladder **1152**. A pressure gauge **1157** is shown to depict pressure on the fluid **1160** exerted by a

pump (not shown) in the system (e.g., the pump **116** of FIG. **1** or the circulating pump **934** of FIG. **9B**).

(69) As depicted in FIGS. **11A-11F**, each bladder **1151** and **1152** is elastic and resilient. That is, each is configured to elastically deform outward in response to pressurized fluid from within, then resiliently return to its initial state when its elastic deformation force exceeds the force that the fluid exerts on its inner surface.

(70) FIG. **11A** depicts an initial equilibrium state in which each bladder **1151** and **1152** is filled with fluid **1160**, but the net pressure between that of the fluid exerted on the bladders **1151** and **1152** and that of the bladders **1151** and **1152** on the fluid is substantially zero.

(71) FIG. **11B** depicts an increase in pressure of the fluid **1160** on the bladders **1151** and **1152** (e.g., as would occur when a pump in the system begins pumping fluid **1160** into the bladders **1151** and **1152**). Because of the flow restrictor **1163**, the first bladder **1151** fills (and expands) before (or to a greater extent than) the filling and expansion of the second bladder **1152**, although, as depicted, some fluid **1160** begins flowing through the flow restrictor **1163**, into the second bladder **1152**, as the first bladder **1151** fills and expands.

(72) FIG. **11C** depicts continued pressurized flow of fluid **1160** into the first bladder **1151** and second bladder **1152**. The first bladder may reach a maximum capacity, after which further expansion or inflation may be prevented (e.g., see FIGS. **16A-16C**, FIGS. **17A-17C** and corresponding description). At this point, all additional fluid **1160** may flow through the first bladder **1151** and flow restrictor **1163**, into the second bladder **1152**, until the second bladder **1152** also reaches its maximum capacity. At this point, no more fluid **1160** may flow into the bladders **1151** and **1152**, but the fluid **1160** therein may be retained in place by pressure exerted on the fluid **1160** by the pump (as depicted in FIG. **11D**).

(73) FIG. **11E** depicts a release of pressure exerted on the fluid **1160** by a pump, such that the fluid **1160** is pushed out by the elastic force of the bladders **1151** and **1152**. As depicted, the flow restrictor **1163** may limit the rate of outflow from the second bladder **1152**, resulting in the first bladder **1151** deflating prior to the second bladder **1152**.

(74) FIG. **11F** depicts a return to an equilibrium state, in which the pressure of the fluid **1160** exerted on the bladders **1151** and **1152** and the elastic deformation force exerted by the bladders **1151** and **1152** on the fluid **1160** substantially balances, and the fluid **1160** present in the bladders **1151** and **1152** remains there. In some implementations, substantially all fluid **1160** present in the bladders **1151** and **1152** may be optionally evacuated (e.g., at the end of a pumping session, in order to reduce weight or bulk of the system).

(75) In some implementations, elasticity of the first bladder **1151** and second bladder **1152** may be the same; in other implementations, elasticity of the first bladder **1151** and second bladder **1152** may be different—such that the inflation and deflation rates are affected (e.g., a less elastic, stiffer, bladder may inflate more slowly and deflate more quickly than a more elastic, less stiff bladder). Moreover, characteristics (e.g., diameter) of the flow restrictor **1163** and/or characteristics of the inlet/outlet **1154** may be “tuned” to control inflation and deflation rates of the bladders **1151** and **1152**, and the relative inflation and deflation rates of the first bladder **1151** and the second bladder **1152**. By changing such design constraints—the elasticity of the first bladder **1151** and second bladder **1152** and the characteristics of the inlet/outlet **1154** and the flow restrictor **1163**—massage sensations produced by the sequential inflation and deflation of the bladders **1151** and **1152** may be controlled. Pump characteristics (e.g., pressure generated, flow capacity, frequency of pressure change) may also be controlled to tailor the resulting massage sensations in a desired manner.

(76) As depicted in FIGS. **11A-11F**, a pump only exerts a positive pressure on the fluid **1160** to inflate the bladders **1151** and **1152**; and the bladders **1151** and **1152** then deflate when the pump stops exerting that pressure on the fluid **1160** and the force of elastic deformation of the bladders **1151** and **1152** pushes fluid **1160** out the inlet/outlet **1154**. In other implementations, the pump may also exert a negative pressure on the fluid **1160** during the deflation stage, to evacuate the fluid

1160 out of the bladders **1151** and **1152** at a faster rate than is possible in the absence of such negative pressure.

(77) FIGS. **12A-12F** depict operation of an exemplary system with a first bladder **1251** and a second bladder **1252** that is also arranged in a “blind channel” manner, but that includes a one-way check valve **1264**, in addition to a flow restrictor **1263** that couples the first bladder **1251** to the second bladder **1252**. A pressure gauge **1257** is again shown to depict pressure on the fluid exerted by a pump in the system (not shown).

(78) FIG. **12A** depicts an initial equilibrium state in which each bladder **1251** and **1252** is filled with fluid **1260**, but the net pressure between that of the fluid exerted on the bladders **1251** and **1252** and that of the bladders **1251** and **1252** on the fluid is substantially zero.

(79) FIG. **12B** depicts an increase in pressure of the fluid **1260** on the bladders **1251** and **1252** (e.g., as would occur when a pump in the system begins pumping fluid **1260** into the bladders **1251** and **1252**). Because of the flow restrictor **1263** and the orientation of the check valve **1264** (opening to flow from the second bladder **1252** to the first bladder **1251**, but not to flow from the first bladder **1251** to the second bladder **1252**), the first bladder **1251** fills (and expands) before (or to a greater extent than) the filling and expansion of the second bladder **1252**, although, as depicted, some fluid **1160** begins flowing through the flow restrictor **1263**, into the second bladder **1252** as the first bladder **1251** fills and expands.

(80) FIG. **12C** depicts continued pressurized flow of fluid **1260** into the first bladder **1251** and second bladder **1252**. As with the implementation depicted in FIGS. **11A-11F**, the first bladder **1251** may reach a maximum capacity, after which further expansion or inflation may be prevented. At this point, all additional fluid **1260** may flow through the first bladder **1251** and flow restrictor **1263**, into the second bladder **1252**, until the second bladder **1252** also reaches its maximum capacity. At this point, no more fluid **1260** may flow into the bladders **1251** and **1252**, but the fluid **1260** therein may be retained in place by pressure exerted on the fluid **1260** by the pump (as depicted in FIG. **12D**).

(81) FIG. **12E** depicts a release of pressure exerted on the fluid **1260** by a pump, such that the fluid **1260** is pushed out by the elastic force of the bladders **1251** and **1252**. As shown, the check valve **1264** opens, allowing a larger flow from second bladder **1252** to the first bladder **1251** than was possible as fluid **1260** was flowing from first bladder **1251** to second bladder **1252**. As depicted, the second bladder **1252** may deflate before the first bladder **1251**, even though the combination of the flow restrictor **1263** and open check valve **1264** may facilitate a similar flow as that facilitated by the inlet/outlet **1254**. In some implementations, this sequence may be facilitated by material properties of the bladders **1251** and **1252**. For example, the bladder **1252** may be less elastic (stiffer) than the first bladder **1252**, causing it to deflate more quickly.

(82) Characteristics of the check valve **1263** may be adjusted, too. For example, a spring force that must be overcome to open the check valve **1263** may be weaker in a first implementation than in a second implementation, resulting in the check valve **1263** opening more in the first implementation than in the second implementation.

(83) FIG. **12F** depicts a return to an equilibrium state, in which relative pressure of the fluid **1260** exerted on the bladders **1251** and **1252** and elastic deformation force exerted by the bladders **1251** and **1252** on the fluid **1260** is substantially zero, and the fluid **1260** present in the bladders **1251** and **1252** remains there. In some implementations, substantially all fluid **1260** present in the bladders **1251** and **1252** may be optionally evacuated (e.g., at the end of a pumping session, in order to reduce weight or bulk of the system).

(84) FIGS. **13A-13F** depict a variation of the implementation depicted in FIGS. **12A-12F**, in which fluid **1360** is pumped through an inlet/outlet **1354**, into a first bladder **1351** and second bladder **1352**, where a flow restrictor **1363** and check valve **1364** are disposed between the first bladder **1351** and second bladder **1352**. A gauge **1357** is again provided as a reference for the pressure of the fluid **1360**.

(85) Operation of the implementation in FIGS. 13A-13F is similar to that shown in FIGS. 12A-12F: pressurized fluid **1360** inflates the first bladder **1351** faster than the second bladder **1352**, in part, because of a flow restrictor **1363** and closed check valve **1364** (FIG. 13A); pressurized fluid **1360** continues to inflate the first bladder **1351** to its maximum capacity (FIG. 13B) and the second bladder **1352** to its maximum capacity (FIG. 13C); and once both bladders **1351** and **1352** are maximally inflated, continued pressure holds the fluid **1360** in place, though no fluid flows in either direction (FIG. 13D).

(86) As depicted in FIG. 13E, a negative pressure may result in the bladders **1351** and **1352** deflating faster than in implementations with zero pressure—that is, a negative pressure (e.g., a pressure below atmospheric pressure) may evacuate the fluid **1360** more quickly than it would be otherwise ejected through only elastic deformation forces exerted by the bladders **1351** and **1352**. As further depicted, depending on the magnitude of the pressure and the characteristics of the first bladder **1351**, the second bladder **1352**, the flow restrictor **1363** and the check valve **1364**, it may be possible to deflate both bladders **1351** and **1352** at substantially the same time.

(87) FIG. 13F depicts a return to equilibrium state, in one implementation.

(88) FIGS. 14A-14F depict another exemplary first bladder **1451** and second bladder **1452** that are arranged in a “circulating” manner—that is, the bladders **1451** and **1452** are configured such that fluid **1460** flows into the first bladder **1451**, through a dedicated inlet line **1455**, and to and through the second bladder **1452**, exiting through a dedicated outlet line **1456**. In some implementations, the inlet line **1455** and the outlet line **1456** may be coupled at a common inlet/outlet **1454**.

(89) As shown, the first bladder **1451** is coupled to the second bladder **1452** through a flow restrictor **1463**. Check valves may be provided—specifically, a first check valve **1465a** that only allows fluid **1460** to move from the inlet line **1455** to the first bladder **1451**, but not from the first bladder **1451** back to the inlet line **1455**; and a second check valve **1465b** that only allows fluid **1460** to move from the outlet line **1456** to the inlet/outlet **1454**, but not from the outlet/inlet **1454** to the outlet line **1456**. This implementation may provide another means for control over the sequence of deflation of the bladders **1451** and **1452**—in particular, providing another manner to control deflation of the second bladder **1452** before deflation of the first bladder **1451**.

(90) Similar to other implementations, pressurized fluid **1460** inflates the first bladder **1451** faster than the second bladder **1452**, in part, because of a flow restrictor **1463** and closed check valve **1465b** (FIG. 14A); and pressurized fluid **1460** continues to inflate the first bladder **1451** to its maximum capacity (FIG. 14B) and the second bladder **1452** to its maximum capacity (FIG. 14C). In some implementations, once both bladders **1451** and **1452** are maximally inflated (depending on the characteristics of the check valves **1465a** and **1465b** (i.e., their inherent spring force that must be overcome to cause them to open)), no additional pressure may be required (see gauge **1457**, depicting zero pressure in FIG. 14D) to maintain the fluid **1460** in place.

(91) FIG. 14E depicts a negative pressure applied at the inlet/outlet **1454**, which causes the fluid **1460** to be withdrawn from the bladders **1452** and **1451** through the outlet **1456** (as shown, the check valve **1465b** opens; but the check valve **1465a** remains closed, causing fluid **1460** to only be evacuated through the outlet line **1456**). As further depicted, a diameter of the outlet line **1456** may allow for greater flow from the second bladder **1452** than that flow restrictor **1463** permits from the first bladder **1451** to the second bladder **1452**—resulting in faster deflation of the second bladder **1452** than the first bladder **1451**.

(92) FIG. 14F depicts a return to an equilibrium state, in one implementation.

(93) FIGS. 15A-15F depict another exemplary first bladder **1551** and second bladder **1552** that are arranged in a “circulating” manner. This implementation includes an inlet/outlet **1554**, a dedicated inlet line **1555**, a dedicated outlet line **1556**, a flow restrictor **1563**, a first check valve **1565a** and second check valve **1565b**. FIGS. 15A and 15F depict equilibrium states, in one implementation; FIGS. 15B and 15C depict sequential filling of the bladders **1551** and **1552** with fluid **1560**; FIG. 15D depicts an equilibrium state in which the bladders **1551** and **1552** are filled to their maximum

capacity. As depicted by FIG. 15E, the position of the check valve **1565b** may facilitate substantially simultaneous deflation of the bladders **1551** and **1552**.

(94) Numerous other configurations are possible, including configurations with more bladders (three, four, five, etc.). By specifically configuring inlet lines, outlet lines, flow restrictors, elasticity of bladders, and, optionally including check valves, it may be possible to precisely stage the inflation and deflation of bladders in a manner that facilitates an effective massage function. By employing a warmed liquid as the working fluid for inflating bladders, a heating function may be combined with a massage function. Moreover, a pump for circulating fluid through the bladders and various inlet/outlet line or inlet and outlet lines, may be adjusted to provide alternating positive and zero or positive, zero and negative pressure to circulate fluid and inflate and deflate bladders in a cyclic manner. In some implementations, such inflating and deflating bladders may be used in combination with various pumping tops or garments to enhance milk production during a pumping operation by mimicking, at least in part, the massage and heat that a nursing infant may otherwise provide.

(95) FIGS. **16A-16C** illustrate one manner in which a maximum capacity, or distension, of a bladder may be limited. As shown, an outer surface of a bladder **1651** may include fibers **1680** (e.g., non-elastic fibers) that are anchored at only certain points. In a deflated state, the fibers **1680** may have some amount of slack, which may be taken up as the bladder is inflated (as depicted in FIGS. **16B** and **16C**). At this point, the fibers **1680** may prevent further inflation of the bladder **1651**—in some implementations, preventing its rupture.

(96) FIGS. **17A-17C** illustrate another manner in which a maximum capacity, or distension, of a bladder may be limited. As shown, an elastic bladder **1751** may be covered by an inelastic material **1781**. When the bladder **1751** is deflated, the inelastic material **1781** may be loose and folded or pleated (FIG. **17A**), such that when the bladder is inflated (FIGS. **17B** and **17C**), the inelastic material **1781** stretches out to its maximum distension—thereby limiting, in some implementations, the distension of the bladder **1751** and preventing its rupture. (In FIGS. **17B** and **17C**, a gap between the bladder **1751** and inelastic material **1781** is exaggerated; but the reader will appreciate that, in practice, no such gap may exist.) In some implementations, another inelastic or noncompliant material **1752** may be disposed on an opposite side as the bladder **1751**, such that the bladder only “inflates” in one direction (e.g., against a user's breast, rather than outward relative to the user).

(97) FIG. **18A** illustrates an exemplary stress-strain curve, in some implementations. Specifically, FIG. **18A** depicts how, as pressure increases (y-axis), a first bladder **1851** may stretch (inflate); and as pressure continues to increase, a second bladder **1851** may stretch (inflate). FIG. **18A** depicts an implementation in which a first bladder **1851** inflates at a first pressure, and a second bladder **1852** only inflates at a second, higher pressure.

(98) In other implementations, as depicted in FIG. **18B**, both bladders **1851** and **1852** may inflate at the same pressure, but the time of inflation may be staggered. That is, at a given minimum pressure (or greater), a first bladder **1851** may first begin inflating, followed (after a short period of time) by a second bladder **1852**. Specific time between inflation of the first bladder **1851** and the second bladder **1852** may depend on various factors, including, for example, any flow restrictors between the bladders **1851** and **1852**, elasticity of the first and second bladders **1851** and **1852** or relative elasticity of one bladder to the other. As outlined elsewhere, various such factors may be “tuned” to sequence inflation or deflation in a desirable manner.

(99) Several implementations have been described with reference to exemplary aspects, but it will be understood by those skilled in the art that various changes may be made and equivalents may be substituted for elements thereof without departing from the contemplated scope. For example, materials with various characteristics such as elasticity, resilience and maximum distension may be employed for bladders; “inelastic” materials may have some elasticity, but far less so than “elastic” materials; maximum capacity of bladders may be limited by the bladder material itself or by

external limiting factors (e.g., fibers, inelastic covering layers); one or more check valves may be employed and at different positions; one or more flow restrictors may be employed; either “blind channel” or “circulating” implementations may be employed; various parameters may be tuned such that bladders inflate in a staged manner and deflate in various staged or substantially simultaneous manners; multiple pumps may be employed (e.g., dedicated pumps for each side, or dedicated pumps for inflating bladders and for circulating warming or cooling fluid); other variations are possible.

(100) In some places, the term “substantially” may mean “completely” or “nearly completely.” For example, the disclosure includes, “In some implementations, the first and second outer expandable elements **300a**, **300b** are configured to fill substantially simultaneously.” In this context, “substantially” is used to mean that the first and second outer expandable elements may be configured to fill simultaneously, or there may be a slight (e.g., a few seconds) delay. For example, the first outer expandable element may begin to fill a few seconds before the second outer expandable element begins to fill (or vice-versa). This slight delay falls under the scope of “substantially simultaneously” as used in this disclosure. In other places, “substantially” (or “about” or “approximately” may mean within 1%, or 5%, or 10%, or 20%, or 50%, or 100% of a nominal value.

(101) Many other variations are possible, and modifications may be made to adapt a particular situation or material to the teachings provided herein without departing from the essential scope thereof. Therefore, it is intended that the scope include all aspects falling within the scope of the appended claims.

Claims

1. A pumping top comprising: a left panel and a right panel, each of the left panel and the right panel being configured to cover a corresponding portion of a side of a wearer's torso and cover and support a corresponding breast of the wearer; wherein each of the left panel and the right panel comprises: (i) a non-expandable reservoir that is configured to contact the corresponding breast of the wearer over substantially 360 degrees adjacent a nipple, the non-expandable reservoir further comprising an aperture which is adapted to facilitate access to the nipple for an external breast pump that is separate and distinct from the pumping top, (ii) an expandable bladder that is separate and distinct from the non-expandable reservoir, configured to be disposed more distally from the nipple than the non-expandable reservoir, fluidly coupled to the non-expandable reservoir and configured to contact and partially surround a corresponding breast of the wearer, and (iii) a line fluidly coupled at one end to one of the expandable bladder or the non-expandable reservoir and at an opposite end to a connector; wherein the connector is configured to couple to a pump that is separate and distinct from the pumping top, which pump alternately pumps warmed liquid into the line and evacuates the liquid from the line.
2. The pumping top of claim 1, wherein each of the right and left panels comprises inner and outer layers, with the line disposed therebetween.
3. The pumping top of claim 1, wherein each connector further comprises a valve to seal off the line when it is not coupled to the pump.
4. The pumping top of claim 1, wherein the cross-sectional shape of the line is flat, with a width dimension being greater than a height dimension.
5. The pumping top of claim 1, wherein the expandable bladder of each side comprises a first expandable bladder and a second expandable bladder, wherein the first expandable bladder is coupled to the line, the second expandable bladder is coupled to a return line, and the first expandable bladder is coupled to the second expandable bladder with a tube.
6. The pumping top of claim 5, wherein the tube is non-expandable.
7. The pumping top of claim 6, wherein the tube has a reduced diameter relative to the first

expandable bladder and the second expandable bladder, such that fluid flow from the first expandable bladder to the second expandable bladder is slowed to cause the first expandable bladder to fill prior to the filling of the second expandable bladder.

8. A pumping top comprising: a left panel and a right panel, each of the left panel and the right panel being configured to cover a corresponding portion of a side of a wearer's torso and cover and support a corresponding breast of the wearer; wherein each of the left panel and the right panel comprises a first expandable bladder and a second expandable bladder, each of the first expandable bladder and the second expandable bladder configured to contact and partially surround the corresponding breast of the wearer and each of the first expandable bladder and the second expandable bladder being fluidly coupled to separate ends of a non-expandable reservoir that is configured to contact the corresponding breast over substantially 360 degrees adjacent a nipple, and a liquid line fluidly coupled at one end to the first expandable bladder and at an opposite end to a connector; wherein each of the left panel and the right panel has an aperture configured to receive an external breast pump that is separate and distinct from the pumping top; and wherein the connector is configured to couple to a circulating pump that is separate and distinct from the pumping top, which circulating pump circulates warmed liquid to and from each liquid line and to and from each non-expandable reservoir.

9. The pumping top of claim 8, wherein the second expandable bladder is coupled to the first expandable bladder with a tube that includes a check valve.

10. The pumping top of claim 9, wherein the tube defines a first channel that includes the check valve and a second channel that is open between the first expandable bladder and the second expandable bladder.

11. The pumping top of claim 8, wherein the second expandable bladder is coupled to the first expandable bladder with a tube that comprises a flow restrictor that constricts flow between the first expandable bladder and the second expandable bladder.

12. The pumping top of claim 11, further comprising a return line that couples the second expandable bladder at an end opposite the tube to the liquid line.

13. The pumping top of claim 12, further comprising a check valve in the return line, in the tube, or in the second expandable bladder.

14. The pumping top of claim 13, wherein an inlet line is disposed between the liquid line and the first expandable bladder, and an inlet check valve is disposed in the inlet line.

15. The pumping top of claim 13, further comprising a second check valve in the return line.

16. The pumping top of claim 8, wherein each of the first expandable bladder and the second expandable bladder comprises means for limiting its distension.

17. The pumping top of claim 8, wherein each of the first expandable bladder and the second expandable bladder comprises non-elastic fibers that are configured to limit distension of the first expandable bladder and the second expandable bladder.
